

Rayleigh Transport on Singular Gram Supports

Kernel Compatibility, Pseudovolume, and Outward Certification

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Abstract

RH-130 showed that positive conditioning floors conceal a genuine semidefinite obstruction: discarded-memory tails can create rank across scales. We develop the singular-Gram theory needed to state the remaining directional argument without regularization. For $G, D \succeq 0$, the global inequality $D \preceq \gamma^2 G$ has a finite constant if and only if $\text{Ker } G \subseteq \text{Ker } D$. Under this compatibility condition its sharp constant is

$$\gamma^2 = \lambda_{\max}(G^{\dagger/2} D G^{\dagger/2}|_{\text{Ran } G}).$$

If $G = A^* A$ has rank r and $R^* R \preceq \gamma^2 G$ with $\gamma < 1$, then the product of the r nonzero singular values satisfies the sharp bound

$$\text{vol}_r(A + R) \geq (1 - \gamma)^r \sqrt{\text{pdet } G}.$$

We also derive an outward support formula: a compressed numerical quotient q with Gram and tail radii r_G, r_D becomes $q + (r_D + q r_G)/(\hat{g} - r_G)$, where $\hat{g}$ is the compressed Gram gap.

A 4,096-instance compatible audit verifies the pseudoinverse formula, its sharp volume example, and all outward bounds; 1,024 injected kernel-leakage examples are all rejected. Applied to RH-130, the theorem finds 54 zero-tail states and 66 full-tail states. The 24 transitions that create tail rank coincide exactly with the 24 infinite multiplicative factors. Singular support is therefore controllable, but only after the full-space kernel gate and the additive birth term are kept explicit.

1 The distinction between a support theorem and a full theorem

Let H be finite dimensional and let $G, D \succeq 0$. The generalized Rayleigh quotient

$$\mathcal{R}_{D,G}(x) = \frac{\langle Dx, x \rangle}{\langle Gx, x \rangle}$$

is defined whenever the denominator is positive. If $G \succ 0$, its supremum is the largest eigenvalue of $G^{-1/2} D G^{-1/2}$. If G is singular, two statements that are often conflated must be separated.

First, on $E = \text{Ran } G = (\text{Ker } G)^\perp$, the compression G_E is positive definite and the quotient is ordinary. Second, an operator inequality on all of H additionally asks that D vanish on $\text{Ker } G$. No matter how small a positive mass on the kernel is, it cannot be dominated by any finite multiple of G . Thus support restriction is stable under a spectral gap, while full-space compatibility is a discontinuous yes/no condition. This is the same rank boundary exposed numerically in RH-130.

Write P for the orthogonal projector onto E and $Q = I - P$. The Moore–Penrose inverse $G^\dagger$ acts as G_E^{-1} on E and as zero on $\text{Ker } G$.

2 Sharp singular Rayleigh theorem

Theorem 1 (Kernel compatibility and sharp constant). *For positive semidefinite G, D the following are equivalent:*

1. *there is a finite c such that $D \preceq cG$ on H ;*
2. *$\text{Ker } G \subseteq \text{Ker } D$;*
3. *$D = PDP$ and $\text{Ran } D \subseteq \text{Ran } G$.*

When they hold, the least constant is

$$c_* = \lambda_{\max}(G^{\dagger/2} D G^{\dagger/2}|_E).$$

Without compatibility, the sharp full-space constant is $+\infty$, although the compressed constant on E remains finite.

Proof. If $D \preceq cG$ and $x \in \text{Ker } G$, positivity gives $0 \leq \langle Dx, x \rangle \leq 0$, hence $D^{1/2}x = 0$ and $Dx = 0$. This proves the kernel inclusion. For self-adjoint operators the inclusion is equivalent to $\text{Ran } D \subseteq \text{Ran } G$ and to $D = PDP$. On E , conjugate $D_E \preceq cG_E$ by $G_E^{-1/2}$. The least c is the largest eigenvalue of the resulting positive matrix. Extending by zero on $\text{Ker } G$ proves the full inequality. If compatibility fails, a kernel vector with positive D -energy makes every finite comparison impossible. $\square$

Theorem 1 is also an immediate finite-dimensional instance of Douglas range factorization [Douglas, 1966]. It supplies an exact meaning for a “relative tail constant” even when the recent action loses rank; no positive floor is required.

Corollary 2 (No small-leakage relaxation). *For every $\varepsilon > 0$ there are $G, D \succeq 0$ with $\|DQ\| = \varepsilon$ such that the support quotient is bounded but the full-space quotient is infinite.*

Proof. Take $G = \text{diag}(1, 0)$ and $D = \text{diag}(c, \varepsilon)$. The support constant is c , while the second coordinate violates every finite full-space inequality. $\square$

This sharp negative statement prevents an outward numerical radius from being silently interpreted as a full-space theorem. One must either prove exact kernel compatibility from the construction or state a theorem only on a certified support.

3 Pseudovolume under a relative tail

Suppose $A, R : H \rightarrow K$ and $G = A^*A$ has rank r . Define

$$\text{vol}_r(A) = \prod_{j=1}^r s_j(A) = \sqrt{\text{pdet}(A^*A)},$$

where only nonzero singular values enter. This is the natural exterior volume on the effective support.

Theorem 3 (Sharp supported-volume transfer). *If*

$$R^*R \preceq \gamma^2 A^*A, \quad 0 \leq \gamma < 1,$$

then $\text{Ker } A \subseteq \text{Ker } R$, $A + R$ has rank r , and

$$\text{vol}_r(A + R) \geq (1 - \gamma)^r \text{vol}_r(A) = (1 - \gamma)^r \sqrt{\text{pdet } G}.$$

The constant is sharp in every rank.

Proof. Theorem 1 first gives $\text{Ker } A \subseteq \text{Ker } R$. Douglas factorization then gives $R = TA$ on the closure of $\text{Ran } A$, with $\|T\| \leq \gamma$. For $y \in \text{Ran } A$, $\|(I + T)y\| \geq (1 - \gamma)\|y\|$. The min-max principle therefore lowers each of the r nonzero singular values by at most the factor $1 - \gamma$; multiplication gives the result. Taking $R = -\gamma A$ gives equality. $\square$

For $r = 4$ this recovers the determinant factor used upstream. For $r < 4$ it does not manufacture a four-volume: it gives the correct r -dimensional exterior statement. In particular, a future varying-rank packet may carry a rigorous pseudovolume while still failing the four-directional Stage A target. Keeping the exterior order visible prevents that logical jump.

4 Outward support certification

Let U be an orthonormal basis for a proposed r -dimensional support and suppose

$$\|G - \hat{G}\| \leq r_G, \quad \|D - \hat{D}\| \leq r_D.$$

Assume the smallest eigenvalue of $\hat{G}_U = U^*\hat{G}U$ is $\hat{g} > r_G$, and let q be the largest generalized eigenvalue of $(U^*\hat{D}U, \hat{G}_U)$.

Proposition 4 (Outward compressed Rayleigh bound). *On UC^r one has*

$$U^*DU \preceq q_{\text{out}} U^*GU, \quad q_{\text{out}} = q + \frac{r_D + qr_G}{\hat{g} - r_G}.$$

Proof. The numerical comparison gives $U^*\hat{D}U \preceq qU^*\hat{G}U$. Hence

$$U^*DU \preceq qU^*GU + (r_D + qr_G)I.$$

Weyl's bound gives $U^*GU \succeq (\hat{g} - r_G)I$, so the identity term is at most $U^*GU/(\hat{g} - r_G)$. Substitution yields the formula. $\square$

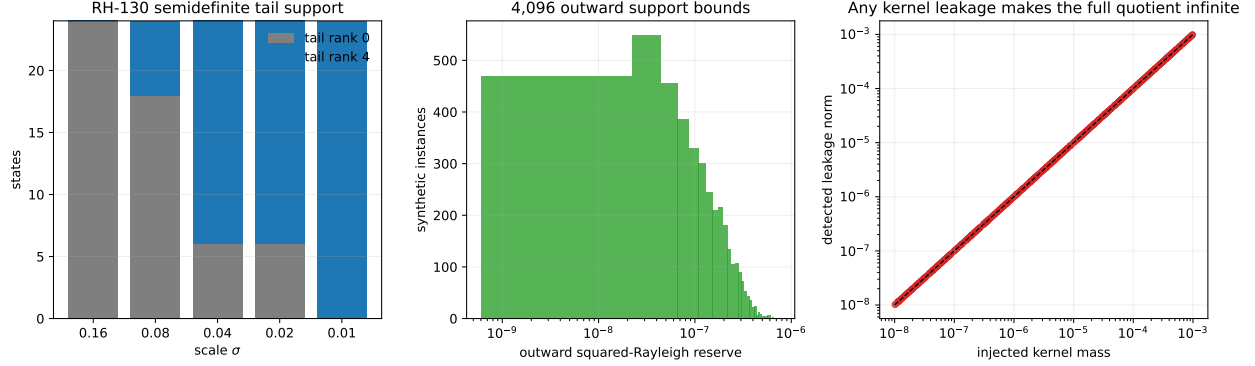


Figure 1: RH-130 tail ranks, outward support reserves, and the discontinuous kernel-leakage obstruction.

The correction is additive in the squared Rayleigh constant and becomes large exactly when the support gap approaches the Gram radius. This is the singular-support analogue of the RH-127 congruence guard. It certifies the compressed theorem only; a separate exact or structural argument is still needed for kernel compatibility.

5 Audit and application to the floor-free packet

We generate 4,096 random positive semidefinite Gramians of ambient dimension seven and ranks one through four. Compatible tails are constructed from a random relative spectrum with prescribed maximum $\gamma^2 < 1$. Every pseudoinverse quotient recovers the prescribed γ to numerical error, the extremal choice $R = -\gamma A$ attains the volume bound, and all 4,096 independently perturbed outward estimates dominate the hidden exact quotient. A second set of 1,024 examples injects kernel mass between 10^{-8} and 10^{-3} ; every example retains a finite compressed quotient and is correctly rejected as an infinite full-space quotient.

The RH-130 archive provides a non-synthetic application. Its 120 states split into 54 states with zero normalized-tail rank and 66 with rank four; no intermediate ranks occur in this frozen packet. At $\sigma = 0.16$ all 24 tails are zero. Nonzero tails enter at later scales and phases. Across the 96 phase-matched adjacent pairs, exactly 24 transitions increase tail rank from zero to four. This count equals the 24 infinite exact-Gram multiplicative factors, case by case. Thus the extended-real minimax result of RH-130 and the kernel-support theorem here diagnose the same structural event from two directions.

The data also clarify what remains viable. The recent-action Gramians in RH-130 are rank four, so their local four-volume does not need a singular replacement. Singularity enters through the tail pencil and through future validated support choices. The next cross-scale construction must transport the old supported tail by a partial isometry and place newly born tail range into an additive forcing block. A post hoc full-rank gauge cannot do this without hiding the rank event.

We have proved the exact singular Rayleigh formula, a sharp pseudovolume theorem, the impossibility of a small-leakage full-space relaxation, and an outward compressed bound. We have not proved a natural all-level support projector, a dynamical partial-isometry

gauge, uniform affine coefficients, a normalized-base liminf, uniform Stage A, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

References

Ronald G. Douglas. On majorization, factorization, and range inclusion of operators on hilbert space. *Proceedings of the American Mathematical Society*, 17:413–415, 1966.